

Background info about mantissa:

(4) Real number can be written as: $x = \pm 0, m \cdot b^e$

$\nearrow$ exponent
 $\downarrow$ mantissa
 $\searrow$ numeration base

Ex: $1234,5678 = 0,12345678 \cdot 10^4$

$0,004371 = 0,4371 \cdot 10^{-2}$

$1101,0011(2) = 0,11010011(2) \cdot 2^4$

Normalised vs not-normalised mantissa:

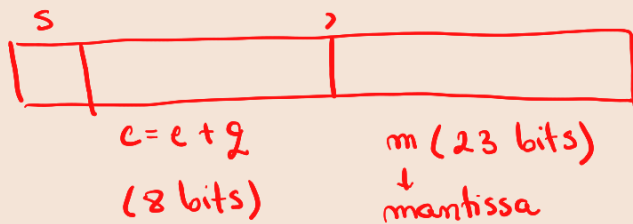
$0,1010011(2) \cdot 2^4 \rightarrow$ subunitary normalised mantissa

$0,001101(2) \cdot 2^{-4} \rightarrow$ subunitary not-normalised mantissa

$\Rightarrow 0,11101(2) \cdot 2^{-6} \Rightarrow$ normalised mantissa

Floating-point representation:

SP = Single Precision



127 (constant)

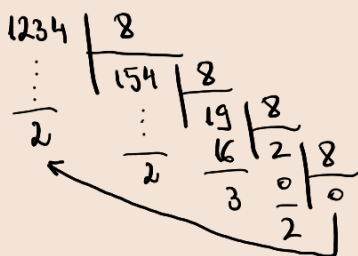
a) $m > 1 \Rightarrow$ add the hidden bit.

b) $m < 1$, normalised.
 $\Rightarrow$ don't add the hidden bit.

① SP, $m > 1$, $x = 1234,56$

$\downarrow$
 floating point, we move the decimal point $\Rightarrow$ add the hidden bit.

Divisions by 8: (how many? until 0)



$\Rightarrow 1234_{(10)} = 02322_{(8)} = 010\ 011\ 010\ 010_{(2)}$

$\downarrow$
 $4 \times 3 = 12$ bits

$8 = 2^3 \Rightarrow$ groups of 3

Multiplications by 8: (how many? 5)

$0,56 \cdot 8 = 4,48$

$0,48 \cdot 8 = 3,84$

$\Rightarrow 0,56_{(10)} = 0,43656_{(2)}$

000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

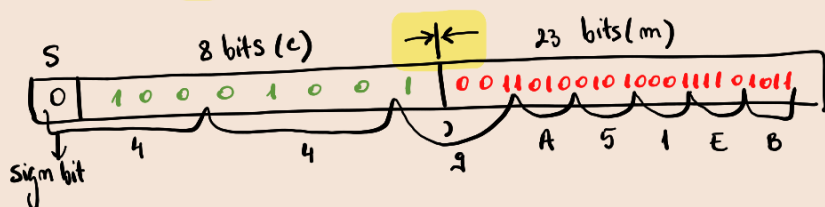
$$\begin{aligned} 0,84 \cdot 8 &= 6,72 \\ 0,72 \cdot 8 &= 5,76 \\ 0,76 \cdot 8 &= 6,08 \end{aligned}$$

$$= 0,100\ 011\ 110\ 101\ 110\ (2)$$

$$5 \times 3 = 15 \text{ bits.}$$

$$\begin{aligned} X = 1234,56 &= 010\ 011\ 010\ 010, 100\ 011\ 110\ 101\ 110\ (2) \\ &= \underbrace{10\ 011\ 010\ 010}_{11 \text{ bits}}, \underbrace{100\ 011\ 110\ 101\ 110}_{15 \text{ bits}}\ (2) \end{aligned}$$

$$\begin{aligned} &\stackrel{m > 1}{=} 1, \underbrace{0011010010}_{10 \text{ bits}} \underbrace{10001111010110}_{15 \text{ bits}}\ (2) \times 2^{10} = e. \\ &\quad \downarrow \\ &\text{hidden bit} \\ &\text{(not represented internally)} \end{aligned}$$



$$\begin{aligned} C = e + g &= 10 + 127 = 137_{(10)} = 2^7 + 2^3 + 2^0 = \underline{1\ 0\ 0\ 0\ 1\ 0\ 0\ 1}\ (2) \\ \downarrow \\ \text{characteristica} &= \text{exponent} + \text{bias}\ (127) \end{aligned}$$

② Floating point, SP, $m < 1$

$$X = -0,12 =$$

Convert into binary:

How many multiplications by 8?

$$\begin{aligned} 0,12 \cdot 8 &= 0,96 \\ 0,96 \cdot 8 &= 7,68 \\ 0,68 \cdot 8 &= 5,44 \end{aligned}$$

$$X = -0,12 = -0,075 \dots (8) = -0, \overbrace{000\ 111\ 101 \dots}^{\text{non-normalised mantissa}}\ (2)$$

Normalise the mantissa:

$$= -0,111\ 101 \dots (2) \times 2^{-3} = e$$

6 more multiplications by 8: (17; $6 \cdot 2 = 12$, $6 \cdot 3 = 18 > 17$)

$$\begin{aligned} 0,44 \cdot 8 &= 3,52 \\ 0,52 \cdot 8 &= 4,16 \\ 0,16 \cdot 8 &= 1,28 \\ 0,28 \cdot 8 &= 2,24 \\ 0,24 \cdot 8 &= 1,92 \end{aligned}$$

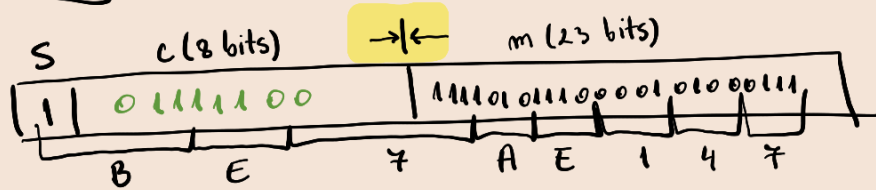
$$0,928 = 7,36 \downarrow$$

$$x = -0,12 = -0,075341217 (8) = -0,00011101010001000111 (2)$$

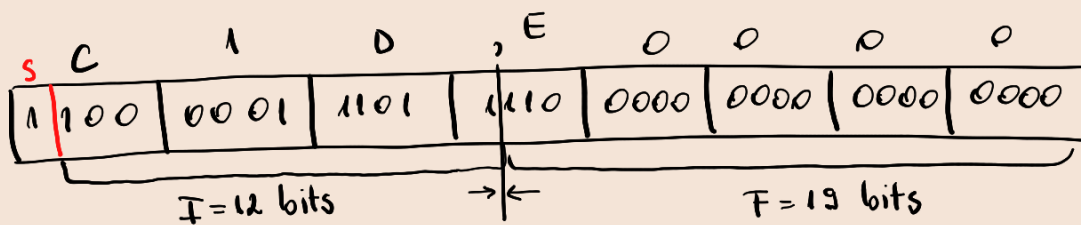
$$= -0,11101010001000111 (2) * 2^{-3} = e$$

normalised subunitary mantissa.

$$[c = e + 9] = -3 + 127 = 124 (10) = 2^6 + 2^5 + 2^4 + 2^3 + 2^2 = 0111100 (2)$$



③ Find the real number x having $CIDE0000 (16)$ as its **fixed-point** representation on 32 bits, $I=12$ bits.



$$x = -100000111011 (2) \quad \begin{array}{l} \text{sums of pw of 2} \\ \text{substitution} \\ \text{source} < \text{dest} \end{array}$$

$$= - (2^0 + 2^1 + 2^3 + 2^4 + 2^5 + 2^{11}), 2^{-1} + 2^{-2}$$

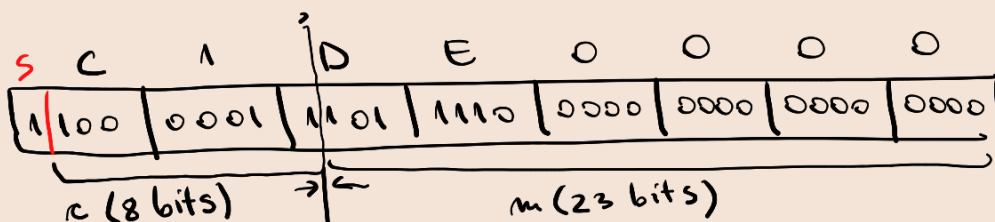
$$= -2107, \frac{1}{2} + \frac{1}{4}$$

$$= -2107,75 (10).$$

④ Find the real number x having $CIDE0000 (16)$ as its **floating-point** representation, SP.
↳ Single Precision.

$$a) m > 1$$

$$b) m < 1, \text{ normalised.}$$



$$c = 10000011 (2) = 2^0 + 2^1 + 2^7 = 131 (10).$$

$$[c = e + 9] = e + 127 = 131 \Rightarrow \underline{e = 4}$$

a) $m > 1 \Rightarrow$ put the sign bit,
 $\Rightarrow$ add the hidden bit.

$$x = -1, m * 2^{\overbrace{4}^{e}} = -1, \overbrace{101111}^{\text{move the "1" 4 positions} \rightarrow} * 2^4$$

$$\downarrow$$

hidden bit

$$= -11011, 11 (2)$$

$$= -(2^0 + 2^1 + 2^3 + 2^4), 2^{-1} + 2^{-2}$$

$$= -(1 + 2 + 8 + 16), 75$$

$$= -27, 75 (10)$$

b) $m < 1 \Rightarrow$ we don't add the hidden bit.

$$x = -0, \overbrace{101111}^{\text{move the "1" 4 positions} \rightarrow} * 2^4 = -1011, 11 (2) = -11, 75 (10)$$

Propositional logic

(1.4) Check the distributivity of $\uparrow$ over $\downarrow \rightarrow$ truth tables

$$\underbrace{p \uparrow (q \downarrow r)}_U \stackrel{?}{=} \underbrace{(p \uparrow q) \downarrow (p \uparrow r)}_V$$

1. Provide theoretical results:

$p \uparrow q := \neg(p \wedge q)$ (nand) is F when all are T

$p \downarrow q := \neg(p \vee q)$ (nor) is T when all are F.

! Two formulas U and V are logically equivalent $\Leftrightarrow$
 $\Leftrightarrow (\forall i, i(U) = i(V)) \rightarrow$ they have identical truth tables.

	p	q	r	$q \downarrow r$	$\underbrace{p \uparrow (q \downarrow r)}_U$	$p \uparrow q$	$p \uparrow r$	$\underbrace{(p \uparrow q) \downarrow (p \uparrow r)}_V$
i1	T	T	T	F	T	F	F	T
i2	T	T	F	F	T	F	T	F
i3	T	F	T	F	T	T	F	F
i4	T	F	F	T	F	T	T	F
i5	F	T	T	F	T	T	T	F
i6	F	T	F	F	T	T	T	F
i7	F	F	T	F	T	T	T	F
i8	F	F	F	T	T	T	T	F

Conclusion: $i_2(U) = T \rightarrow i_1(U) + i_3(U) + \dots + i_8(U) \neq V$

$i_2(V) = F \quad \neg \Rightarrow i_2(U) \neq i_2(V) \Rightarrow U \neq V$
 $\Rightarrow$ Distributivity of $\uparrow$ over $\downarrow$ is not a valid law.

(2.4) Decide the type of the formula:

$$U_4 = \underbrace{\neg(\neg p \wedge q) \vee r}_A \rightarrow \underbrace{p \wedge (\neg q \vee r)}_B$$

1. Provide theoretical results:

order of precedence: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$

• U is consistent if $\exists i, i(U) = T$
 $i \rightarrow$ model of U

• U is valid if $(\forall i), i(U) = T$
 (tautology)
 $\models U$

• U is inconsistent if $(\forall i), i(U) = F$

• U is contingent if $\exists i, i(U) = T$ (model)
 and
 $\exists j, j(U) = F$ (anti-model)

p	q	r	$\neg p$	$\neg q$	$\neg p \wedge q$	$\neg q \vee r$	$\neg(\neg p \wedge q)$	A	B	$A \rightarrow B$
T	T	T	F	F	F	T	T	T	T	T
T	T	F	F	F	F	F	T	T	F	F
T	F	T	F	T	F	T	T	T	T	T
T	F	F	F	T	F	T	T	T	T	T
F	T	T	T	F	T	T	F	T	F	F
F	T	F	T	F	T	F	F	F	F	T
F	F	T	T	T	F	T	T	T	F	F
F	F	F	T	T	F	T	T	T	F	F

Conclusion: U_4 contingent $\rightarrow$ 4 models: $(i_1, i_3, i_4, i_6)(U_4) = T$
 $\rightarrow$ 4 anti-models: $(i_2, i_5, i_7, i_8)(U_4) = F$

$$i_1: \{p, q, r\} \rightarrow \{T, F\}$$

$i_1(p) = T, i_1(q) = T, i_1(r) = T, i_1(U_4) = T \Rightarrow$ model.

$$iz: \{P, Q, R\} \rightarrow \{T, F\}$$

$$iz(p) = T, iz(q) = T, iz(r) = F, iz(U_1) = \textcolor{red}{F} \Rightarrow \text{anti-model.}$$